

DHRUV RAJVANSH

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EDUCATION

University of Southern California

Los Angeles, CA, US

Master of Science, Computer Science – 3.67 / 4 GPA

August 2024 - May 2026

Courses: Analysis of Algorithm, 3D Graphics & Rendering, Computer Animation & Simulation, DB System, Game Engine Development

Pandit Deendayal Energy University

Gandhinagar, Gujarat, India

Bachelor of Technology, Computer Engineering – 3.86 / 4 GPA

June 2020 - April 2024

Courses: Operating Systems, Software Engineering, Big Data Analytics, Cloud Computing, Web Development, Computer Vision, ML

EXPERIENCE

Software Engineer | USC Institute for Creative Technologies, Los Angeles, CA

July 2025 - December 2025

- **Military Simulation Tool:** Built an interactive mission planning tool in Unity (C#, WebGL), integrating Cesium heightmaps to render high-fidelity 3D operational environments.
- **VR Training Migration:** Ported the VITA training system from desktop to Meta Quest 3 using C# and Meta SDK, adding passthrough, room-scale locomotion, and an MQTT-based data pipeline.
- **AI Personality System:** Designed a 3-disposition virtual human framework (neutral/angry/soft), reducing initialization time via optimized prefab architecture and cutscene generation.

Software Engineer Intern | IDZ Digital Private Limited, India

January 2024 - July 2024

- **Mobile Game Prototyping:** Prototyped 13 interactive Unity mobile applications (Planet, Car Stacking Snake, Mandala design) meeting all deadlines under strict time constraints.
- **Runtime Performance Optimization:** Boosted frame rate by 45% and improved responsiveness by deploying Unity memory management strategies, streamlining asset workflows, and reducing CPU bottlenecks through architectural improvements.

Software Engineer Intern | Vyorius Drone Private Limited, India

June 2023 - August 2023

- **Multiplayer Networking Architecture:** Coded real-time communication layers in C#, enabling precise state tracking and real-time multi-user synchronization for low-latency remote drone operation across distributed systems.
- **Network Latency Optimization:** Improved UI flows including lobby panels and score tracking systems, and redesigned networking protocols and state synchronization, reducing network latency by 40%.
- **Mirror API Integration:** Leveraged Mirror API to manage distributed state-sync events, establishing session instancing, secure authority handling, and reliable remote interaction for scalable multiplayer environments.

TECHNICAL PROJECTS & RESEARCH

Interactive WebXR Visualization Platform - Three.js, WebGL, HTML5, JavaScript, GLSL Shaders.

- Engineered a VR teleoperation prototype with CCD-based inverse kinematics driven by hand tracking, mapping pose to joint targets for robot arm control.
- Designed a modular World Manager supporting dynamic loading and isolation of 5 independent 3D environments for scalable multi-context operator interfaces.
- Built a chunked HTTP byte-range video streaming pipeline projecting equirectangular frames onto spherical geometry for 360° panoramic playback at 60 FPS, using object pooling to eliminate GC overhead.

Prime Engine - Real-Time Graphics Engine Development: C++, DirectX 11, OpenGL, HLSL, Lua, Python

- Engineered a distance-based mesh LOD system, dynamically swapping high-fidelity meshes with low-poly proxies to reduce draw calls and vertex overhead.
- Implemented temporal skeletal animation optimization modulating skinning update frequency by camera distance, reducing CPU cycles while preserving motion sync.
- Architected modular C++ components for an entity-component system with spatial state queries and real-time debug visualization.

SKILLS

Languages: C++, C#, Python, Java, JavaScript, GLSL, CUDA, SQL

Engines & Graphics: Unity, Unreal Engine, OpenGL, DirectX 11, Vulkan, WebGL, WebXR API, Three.js, Meta SDK, OpenXR

Tools & DevOps: Git, GitHub, Perforce, Visual Studio, VS Code, AWS, Azure, Docker, GitHub Actions, Postman

Core CS: Real-time Systems, Multithreading, Distributed Systems, Performance Optimization, System Design, Data Structures & Algorithms